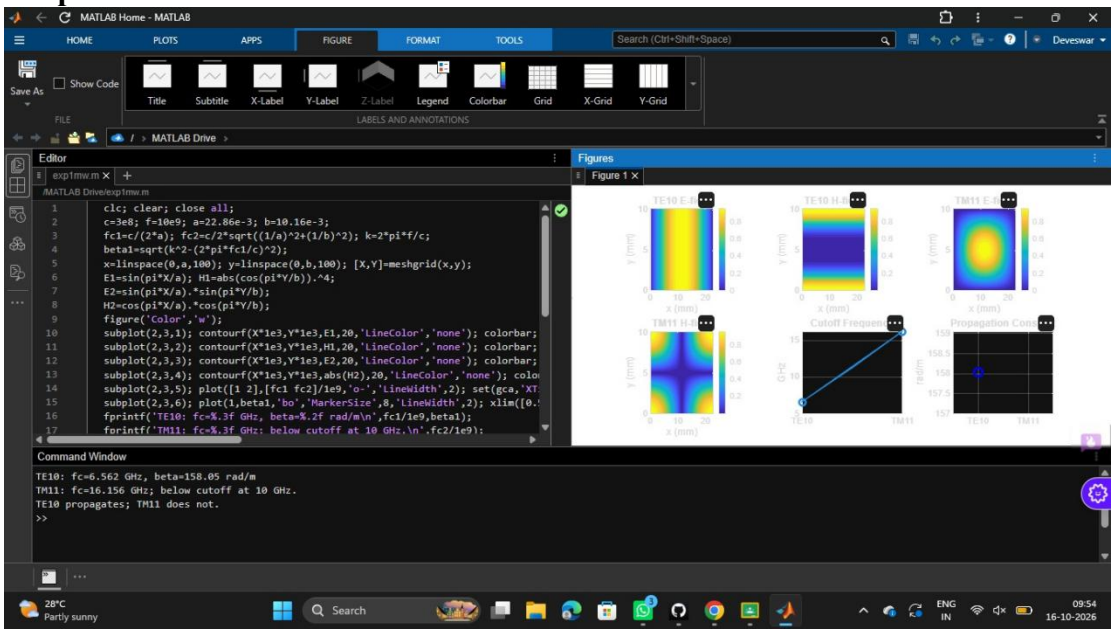
Exp no:1

Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

Code

```
clc; clear; close all;
c=3e8; f=10e9; a=22.86e-3; b=10.16e-3;
fc1=c/(2*a); fc2=c/2*sqrt((1/a)^2+(1/b)^2); k=2*pi*f/c;
beta1=sqrt(k^2-(2*pi*fc1/c)^2);
x=linspace(0,a,100); y=linspace(0,b,100); [X,Y]=meshgrid(x,y);
E1=sin(pi*X/a); H1=abs(cos(pi*Y/b)).^4;
E2=sin(pi*X/a).*sin(pi*Y/b);
H2=cos(pi*X/a).*cos(pi*Y/b);
figure('Color','w');
subplot(2,3,1); contourf(X*1e3,Y*1e3,E1,20,'LineColor','none'); colorbar; title('TE10
E-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,3,2); contourf(X*1e3,Y*1e3,H1,20,'LineColor','none'); colorbar; title('TE10
H-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,3,3); contourf(X*1e3,Y*1e3,E2,20,'LineColor','none'); colorbar; title('TM11
E-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,3,4); contourf(X*1e3,Y*1e3,abs(H2),20,'LineColor','none'); colorbar;
title('TM11 H-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,3,5); plot([1 2],[fc1 fc2]/1e9,'o-','LineWidth',2); set(gca,'XTick',[1
2],'XTickLabel',{'TE10','TM11'}); ylabel('GHz'); title('Cutoff Frequency'); grid on;
subplot(2,3,6); plot(1,beta1,'bo','MarkerSize',8,'LineWidth',2); xlim([0.5 2.5]);
set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); ylabel('rad/m'); title('Propagation
Constant'); grid on;
fprintf('TE10: fc=%.3f GHz, beta=%.2f rad/m\n',fc1/1e9,beta1);
fprintf('TM11: fc=%.3f GHz; below cutoff at 10 GHz.\n',fc2/1e9);
fprintf('TE10 propagates; TM11 does not.\n');
```

Output


MATLAB Home - MATLAB
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Deveswar
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Editor
exp1mw.m
/MATLAB Drive/exp1mw.m
clc; clear; close all;
c=3e8; f=10e9; a=22.86e-3; b=10.16e-3;
fc1=c/(2*a); fc2=c/2*sqrt((1/a)^2+(1/b)^2); k=2*pi*f/c;
beta1=sqrt(k^2-(2*pi*fc1/c)^2);
x=linspace(0,a,100); y=linspace(0,b,100); [X,Y]=meshgrid(x,y);
E1=sin(pi*X/a); H1=abs(cos(pi*Y/b)).^4;
E2=sin(pi*X/a).*sin(pi*Y/b);
H2=cos(pi*X/a).*cos(pi*Y/b);
figure('Color','w');
subplot(2,3,1); contourf(X*1e3,Y*1e3,E1,20,'LineColor','none'); colorbar;
subplot(2,3,2); contourf(X*1e3,Y*1e3,H1,20,'LineColor','none'); colorbar;
subplot(2,3,3); contourf(X*1e3,Y*1e3,E2,20,'LineColor','none'); colorbar;
subplot(2,3,4); contourf(X*1e3,Y*1e3,abs(H2),20,'LineColor','none'); colo
subplot(2,3,5); plot([1 2],[fc1 fc2]/1e9,'o-','LineWidth',2); set(gca,'XT
subplot(2,3,6); plot(1,beta1,'bo','MarkerSize',8,'LineWidth',2); xlim([0.
fprintf('TE10: fc=%.3f GHz, beta=%.2f rad/m\n',fc1/1e9,beta1);
fprintf('TM11: fc=%.3f GHz; below cutoff at 10 GHz.\n',fc2/1e9);
Figures
Figure 1
Command Window
TE10: fc=6.562 GHz, beta=158.05 rad/m
TM11: fc=16.156 GHz; below cutoff at 10 GHz.
TE10 propagates; TM11 does not.
>>
28°C
Partly sunny
Search
ENG
IN
09:54
16-10-2026